FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – II
PAPER –2
TEST DATE: 01-05-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. Since P and Q are points on the same wavefront, they are in the same phase. Therefore, the path difference at point P between the ray BP and the reflected ray OP is

Δ = QO + OP

Now, in triangle POR, OP = $\frac{PR}{\cos\theta} = \frac{d}{\cos\theta}$

Also in triangle QOP, QO = OP sin(90° – 2θ) = OP cos 2θ.

$\therefore$ Path difference, Δ = OP cos 2θ + OP

= OP(1 + cos 2θ)

= 2OP $\cos^2\theta$

= $\frac{2d}{\cos\theta}\cdot\cos^2\theta$

= 2d cos θ

Since, there is sudden path change of $\frac{\lambda}{2}$ due to reflection

$$\Delta = 2d\cos\theta + \frac{\lambda}{2} = n\lambda$$

$$2d\cos\theta = \left(n - \frac{1}{2}\right)\lambda$$

$$\cos\theta = \frac{2n-1}{4d}\lambda$$

n = 1

$$\cos\theta = \frac{\lambda}{4d}$$

2. D

Sol. Angular magnification $m = \frac{f_0}{f_e} = \frac{150}{5} = 30.$

$$\frac{\tan\beta}{\tan\alpha} = 30$$

$\tan\beta = 30 \tan\alpha = \left(\frac{50}{1000}\right) \times 30 = \frac{3}{2}$

$$\beta = \tan^{-1}\left(\frac{3}{2}\right)$$

$\theta \simeq 60°.$

3. C

Sol. $B = \frac{\mu_0 I}{2c} \otimes - \frac{\mu_0 I}{2b}\frac{3}{4} \odot + \frac{\mu_0 I}{2a}\left(\frac{3}{4}\right) \otimes$

$B = \frac{\mu_0 I}{2}\left(\frac{1}{c} - \frac{3}{4b} + \frac{3}{4a}\right)$

$B = 0$

$\Rightarrow \quad \frac{1}{c} = \frac{3}{4}\left(\frac{1}{b} - \frac{1}{a}\right)$

If $c = 2a$

$a = \frac{5}{3}b$

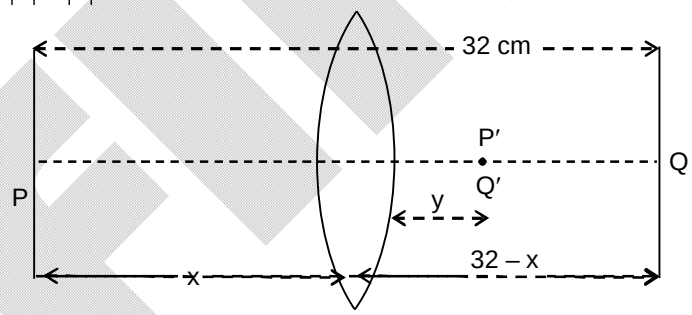
4. A

Sol. $2 \times a + 1 \times 2 + 1 \times 4 = 0.$

5. AD

Sol. Since images are formed at the same place. So one image must be real and other must be virtual

$|x| + |y| = 32$



For P : $\frac{1}{y} - \frac{1}{-x} = \frac{1}{15} \Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{15}$ …(i)

For Q : $\frac{1}{-y} + \frac{1}{38 - x} = \frac{1}{15}$ …(ii)

From (i) and (ii) x = 12, 20 cm

6. ABCD

Sol. Net force on 2 kg $\le$ 8 N
Net force on 4 kg $\le$ 20 N
Net force on 5 kg $\le$ 30 N

7. BCD

Sol. Sign between x and t is $\oplus$ hence wave is propagating in negative x direction.

$$\text{wave velocity} = \frac{\text{coefficient of } t}{\text{coefficient of } x} = \frac{3}{1}$$

Max value of y is amplitude, y has maximum value when denominator is least i.e.
when $(x+3t)^2$ is zero

$$y = 3m$$

shape of pulse is given by $y = \frac{6}{2+x^2}$ $(t=0)$

shape of pulse is symmetrical about y-axis.

SECTION – B

8. 5

Sol. $\frac{K \times 1 \times 27}{9 \times L} = PAV$ …(1) (V is the velocity of piston)

$PA\frac{L}{2} = nR \times 300$ …(2)

Hence, $\frac{K \times 3}{L} = \frac{2nR \times 300\ V}{L}$

$\Rightarrow \quad V = \frac{K}{100\,R}$

$\frac{P}{20} = \frac{100}{20} = 5.$

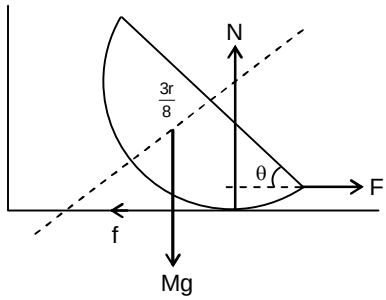
9. 3

Sol. V = constant
f = F = μN = μmg
Torque about point of contact.
Torque due to f and N will be zero.

$(r - r\sin\theta)F = Mg\frac{3r}{8}\sin\theta$

$\Rightarrow \quad \sin\theta = \frac{8\mu}{3+8\mu}$

n = 3.



10. 2

Sol. $t_1 = RC,\ t_2 = \frac{L}{R} \quad \Rightarrow\ LC = t_1t_2 = 0.1$

$T = 2\pi\sqrt{LC} = 2\pi\sqrt{\frac{1}{10}} \simeq 2$ Sec.

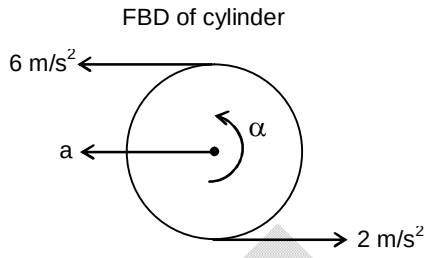
11. 1

Sol. $R\alpha - a = 2$

$a + R\alpha = 6$

$2Rr = 8$

$\alpha = 1$ rad/sec^2



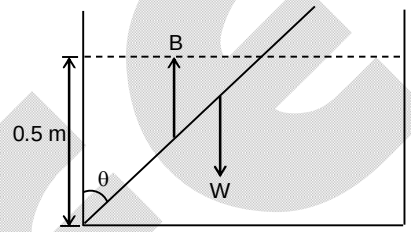
12. 2

Sol. Torque at about hinged point

$$W \times \frac{1}{2}\sin\theta = B \times \frac{0.5\sec\theta\ \sin\theta}{2}$$

$$A\ell dg \times \frac{1}{2}\sin\theta = (A \times 00.5\sec\theta\ \rho g) \times \frac{0.5\sec\theta\sin\theta}{2}$$

$$\sec^2\theta = 2 \quad \Rightarrow \quad \frac{1}{\cos^2\theta} = 2 \quad \Rightarrow \ 2.$$



13. 8

Sol. The emf induced in the sails = B(2r)v

Resistance = $\frac{R}{4}$

$$\Rightarrow \ i = \frac{B(2r)v}{R/4} = \frac{8Brv}{R}$$

$\Rightarrow$ n = 8.

SECTION – C

14. 80.00

Sol. Photo emission will stop if $\lambda = \lambda_0 = \frac{hc}{\phi}$

$$3000 + 40t_0 = \frac{12400}{2},\ t_0 = 80\ \text{s}$$

15. 3.71

Sol. Rate of photo emission at time *t*

$$\frac{dN}{dt} = \frac{1}{100} \times \frac{100}{19840 \times 10^{-19}}[3000 + 40t]$$

$$N = \frac{1}{19840 \times 10^{-19}}[3000t + 20t^2]_0^{80}$$

$$N = 18.55 \times 10^{19} = 1.855 \times 10^{20}$$

This is the number of photoelectron ejected in 1 blinking.

In 1 hr, total number of photoelectron ejected is = $1.855 \times 10^{20} \times 20 = 3.71 \times 10^{21}$

16. 700.00

Sol. Net q charge goes from battery.

So applying Kirchhoff's law $200-\frac{600+q}{6}-\frac{150+q}{3}=0$

q = 100 μC

charge on 6μF is 700.

17. 250.00

Sol. Charge on 3 μC is 250 μC

FIITJEE

Chemistry

PART – II

SECTION – A

18. D

Sol. For n^{th} order

$$t_{1/2} = \frac{1}{k(n-1)}\left[\frac{1}{\left[\frac{C_0}{2}\right]^{n-1}} - \frac{1}{[C_0]^{n-1}}\right]$$

$$\frac{2-\sqrt{2}}{k} C_0^{1/2} = \frac{1}{k}(1-n)(C_0)^{1-n}$$

$$\frac{1}{2} = 1-n$$

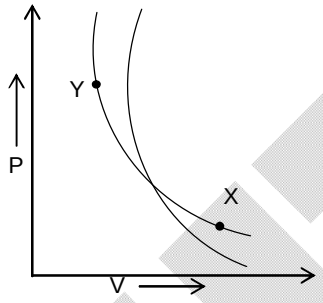
$n = 0.5$

19. B

Sol. Antiaromatic

20. C

Sol.



The curve with points X & Y is for ideal behaviour of the gas, therefore compressibility factor will be 1 from X to Y.

21. B

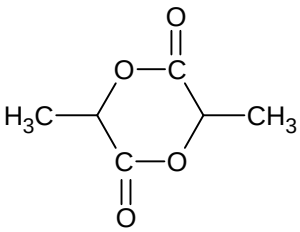
Sol. $6\overset{+4}{Xe}F_4 + 12H_2O \longrightarrow 4\overset{0}{Xe} + 2\overset{+6}{Xe}O_3 + 24HF + 3O_2$

22. ABCD

Sol. (A)The products are $CH_3CH_2CH\begin{matrix}CN\\OH\end{matrix}$ (optical isomerism)

(B) $CH_3CH = N - OH$ (geometrical isomerism).

(C) $CH_3\overset{Cl}{\underset{|}{C}}HCOOH$ (optical isomerism)

(D)  (geometrical isomerism)

23. ABC

Sol. The precipitate is $Fe_4[Fe(CN)_6]_3$.

24. BC

Sol. $\underset{(s)}{2C} + \underset{(g)}{3H_2} \rightarrow \underset{(g)}{C_2H_6}$

$\Delta H_f^0 = \left[2 \times \Delta H_{sub}^0(C) + 3 \times B.E(H-H)\right] - \left[B.E.(C-C) + 6 \times B.E(C-H)\right]$

$\Rightarrow -85 = \left[(2 \times 718) + (3 \times 436)\right] - (x + 6y)$

$\therefore x + 6y = 2829 \quad \ldots\ldots 1$

Similarly for $C_3H_8(g)$

$2x + 8y = 4002 \quad \ldots\ldots 2$

Solving (1) & (2), x = 345

y = 414

SECTION – B

25. 400

Sol. $K_{P_1} = p_{SO_2} \times p_{SO_3} = 900$

$\therefore p_{SO_2} = p_{SO_3} = 30\,\text{cm of Hg}$

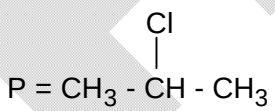
$K_{P_2} = \dfrac{p_{SO_3}^2}{p_{SO_2}^2 \times p_{O_2}} = \dfrac{30}{30 \times p_{O_2}} = 25 \times 10^{-4}$

$\therefore p_{O_2} = \dfrac{1}{25 \times 10^{-4}} = 400\ \text{cm of Hg}$

26. 138

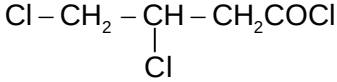
Sol. P = K_2CO_3, Q = O_2, R = KOH, S = H_2O_2. Heaviest product is K_2CO_3.

27. 157

Sol. 

P = $CH_3 - \underset{\displaystyle Cl}{\underset{|}{CH}} - CH_3$

28. 702

Sol. The product is 

$Cl - CH_2 - \underset{\displaystyle Cl}{\underset{|}{CH}} - CH_2COCl$

y = 175.5 g/mol

$\therefore 4y = 702$

29. 453

Sol. The complex is $[Cr(NH_3)_4Cl_2]Cl$.

$x = 226.50$

$\therefore 2x = 453$

30. 4606

Sol. $W = -2.303nRT \log \frac{V_2}{V_1} = -2.303 \times 1 \times 2 \times 1000 \times \log \frac{1/10}{1} = 4606$ cal

SECTION – C

31. 0.50

Sol. $D = \frac{\text{mol.mass of } SO_3}{2} = \frac{80}{2} = 40$

32. 0.67

Sol. $\alpha = \frac{40-30}{(1.5-1)30} = \frac{10}{15} = 0.666 \approx 0.67$

33. 10.62

Sol. $CH_3NH_2(aq) + HCl(aq) \rightleftharpoons CH_3NH_3^+(aq) + Cl^-(aq)$

Mill equ. before reac^n	$200 \times 0.4 = 80$	$200 \times 0.2 = 40$	0	0
Milli equ. after reac^n	$80 - 40 = 40$	$40 - 40 = 0$	40	40

The mixture behaves as a basic buffer

$$pOH = p^{K_b} + \log \frac{\left[CH_3NH_3^+\right]}{\left[CH_3NH_2\right]}$$

$$= 3.38 + \log \frac{40}{40} = 3.38$$

$\therefore$ pH = 14 – pOH = 14 – 3.38 = 10.62

34. 0.10

Sol. $[Cl^-] = \frac{\text{Milli equ.}}{\text{Total volume}} = \frac{40}{200+200} = \frac{40}{400} = 0.1\,M$

Mathematics

PART – III

SECTION – A

35. B

Sol. $S=\frac{1}{1}-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+.....-\frac{1}{2022}+\frac{1}{2023}$

$=\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+.....+\frac{1}{2022}+\frac{1}{2023}$

$-2\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{6}+\frac{1}{8}+.....+\frac{1}{2022}\right)$

$=\sum_{r=1}^{2023}\frac{1}{r}-2\sum_{r=1}^{1011}\frac{1}{2r}=\sum_{r=1}^{2023}\frac{1}{r}-\sum_{r=1}^{1011}\frac{1}{r}=\sum_{1012}^{2023}\frac{1}{r}.$

36. A

Sol. $\operatorname{sgn}\left(\sin^{-1}\left[\frac{\pi x}{6}\right]\right)=1\Rightarrow \sin^{-1}\left[\frac{\pi x}{6}\right]>0\Rightarrow\left[\frac{\pi x}{6}\right]=1\Rightarrow \frac{6}{\pi}\le x<\frac{12}{\pi}.$

37. A

Sol. Line $(x,y,z)=(2+\lambda,-2-3\lambda,5+2\lambda)$

Point $P\equiv(9,-23,19)$ (by solving the line and plane equations)

Point $Q\equiv(0,4,1)$ by solving the line and $x=0$ (YZ – plane)

$PQ=9\sqrt{14}$.

38. D

Sol. $\sum f_i=20$

Median $M=\frac{x_{10}+x_{11}}{2}=\frac{5+5}{2}=5$

M.D. (M) $=\frac{\sum f_i|x_i-M|}{\sum f_i}=\frac{48}{20}=2.4$.

39. BD

Sol. Vector along line $L=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&1&4\\3&2&-2\end{vmatrix}=(2\hat{i}+\hat{j}+4\hat{k})\times(3\hat{i}+2\hat{j}-2\hat{k})$

$=-10\hat{i}+16\hat{j}+\hat{k}$

Line L : $\frac{x-1}{-10}=\frac{y-4}{16}=\frac{z-3}{1}=\lambda$ (say)

$Q\equiv(1-10\lambda,4+16\lambda,3+\lambda);\ P\equiv(1,4,3)$

$(PQ)^2=357\Rightarrow\lambda=1,-1\Rightarrow Q\equiv(-9,20,4)$ or (11, –12, 2).

40. ACD

Sol. $-1 \le \sin x, \cos 2y \le 1 \Rightarrow -1 \le \sin x . \cos 2y \le 1$

$\Rightarrow -1 \le \left(a^2 - 1\right)^2 + 1 \le 1 \Rightarrow a^2 - 1 = 0 \Rightarrow a = \pm 1$ .....(1)

Also $-1 \le \cos x \sin 2y \le 1 \Rightarrow -1 \le a + 1 \le 1 \Rightarrow -2 \le a \le 0$ .....(2)

Hence $a = -1$. So, $\sin x . \cos 2y = 1$ and $\cos x . \sin 2y = 0$

Verify the options.

41. BD

Sol. $\lim\limits_{x \to 0} f(x) = \lim\limits_{x \to 0} P\left(\frac{1}{x^3}\right) e^{-\frac{1}{x^4}} = \lim\limits_{t \to \infty} \left(\frac{P\left(t^3\right)}{e^{t^4}}\right) = 0$.

$f'(0) = \lim\limits_{h \to 0} \frac{P\left(\frac{1}{h^3}\right)}{h e^{\frac{1}{h^4}}} = \lim\limits_{t \to \infty} \left(\frac{tP\left(t^3\right)}{e^{t^4}}\right) = 0$.

SECTION – B

42. 351

Sol. $f(x) = x + \sqrt{1 - x^2}$

$D_f = [-1, 1]$, Range = $\left[-1, \sqrt{2}\right]$

Number of integers in the range = 3, i.e. –1, 0, 1.

43. 2

Sol. $L = \lim\limits_{x \to 0} \frac{\int_0^{\ell n(1+x)} (1 - \tan 2y)^{\frac{1}{y}} dy}{\sin x}$ $\left(\frac{0}{0} \text{ form}\right)$ (L' Hospital Rule)

$= \lim\limits_{x \to 0} \frac{\left(1 - \tan\left(2\,\ell n(1+x)\right)\right)^{\frac{1}{\ell n(1+x)}}}{(\cos x)(1+x)}$ (1^∞ from, numerator)

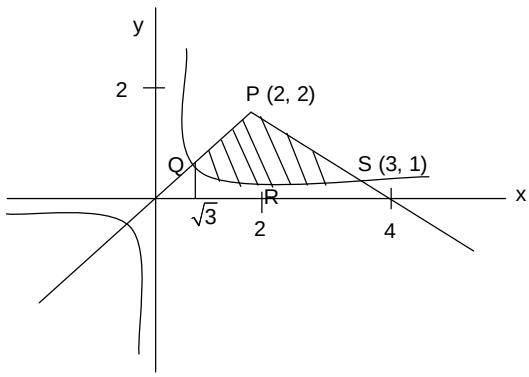
$= \exp\left(\lim\limits_{x \to 0} \frac{-\tan 2\left(\log(1+x)\right)}{\log(1+x)}\right) = e^{-2}$.

44. 4

Sol. $y = \begin{cases} x, & x < 2 \\ 4 - x, & x \ge 2 \end{cases}$

Required area of PQRS

$= \int_{\sqrt{3}}^{2} \left(x - \frac{3}{x}\right) dx + \int_{2}^{3} \left(4 - x - \frac{3}{x}\right) dx = \frac{4 - 3\ell n 3}{2}$



45. 2

Sol. Normal at P (x, y) : $= Y - y = (X - x)\left(-\frac{1}{y'}\right)$

Its x – intercept $= x + yy'$ (Put $Y = 0$)

Mid – point $M \equiv \left(\frac{x + yy' + x}{2}, \frac{y + 0}{2}\right) \equiv \left(\frac{2x + yy'}{2}, \frac{y}{2}\right)$

Then $2\left(\frac{y}{2}\right)^2 = \frac{2x + yy'}{2} \Rightarrow yy' + 2x = y^2 \Rightarrow \frac{1}{2}\frac{dt}{dx} + 2x = t$ (Let $t = y^2$)

$\Rightarrow \frac{dt}{dx} - 2t = -4x$ (Linear DE in t).

On solving, $t = y^2 = 2x + 1 - e^{2x}$.

46. 7

Sol. $\frac{x^2}{2} - \frac{y^2}{3} = 1$ (Hyperbola)

$y = mx \pm \sqrt{2m^2 - 3}$ (tangent)

$\beta = m\alpha \pm \sqrt{2m^2 - 3} \Rightarrow m^2(\alpha^2 - 2) - 2\alpha\beta m + \beta^2 + 3 = 0$ (Roots m_1, m_2)

$m_1 m_2 = \frac{\beta^2 + 3}{\alpha^2 - 2} = 2 = \tan\theta \tan\phi$ (given)

$\Rightarrow 2\alpha^2 - \beta^2 = 7$.

47. 0

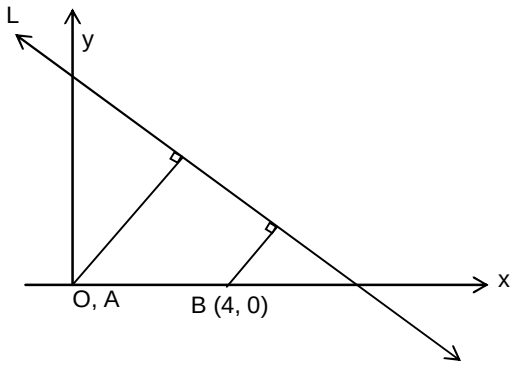
Sol. Family of lines are concurrent at (0, 2) which is same as the focus of the parabola. So, the shortest intercept is the latus rectum which is parallel to x – axis. Therefore, the slope is zero.

SECTION – C

48. 0.37

(Range 0.36 – 0.38)

Sol. Let L be $ax + by = 1$, Then



$p_1 = \frac{1}{\sqrt{a^2 + b^2}}$, $p_2 = \frac{1 - 4a}{\sqrt{a^2 + b^2}}$ and

$A \equiv (0, 0), B \equiv (4, 0)$.

$(1 - 4a > 0$ as A and B are on same side of L).

$p_1 + 3p_2 = k \Rightarrow \frac{1}{\sqrt{a^2 + b^2}} + \frac{3(1 - 4a)}{\sqrt{a^2 + b^2}} = k$

$\Rightarrow \frac{4 - 12a}{\sqrt{a^2 + b^2}} = k \Rightarrow \frac{1 - 3a}{\sqrt{a^2 + b^2}} = \frac{k}{4}$.

So the line $ax + by = 1$ touches the circle with centre (–3, 0).

Circle : $(x + 3)^2 + y^2 = \left(\frac{k}{4}\right)^2$.

49. 1.40
Range (1.39 – 1.41)

Sol. $A(-2, 0), B(2,0)$. Let L be $ax + by - 1 = 0$

$p_1 = \frac{|-2a-1|}{\sqrt{a^2+b^2}}, p_2 = \frac{|2a-1|}{\sqrt{a^2+b^2}}$

A and B are on same side of L so $-2a-1$ and $2a-1$ are of same signs.

$p_1 + 3p_2 = k \Rightarrow \frac{|(-2a-1)+3(2a-1)|}{\sqrt{a^2+b^2}} = k \Rightarrow \frac{|4a-4|}{\sqrt{a^2+b^2}} = k$

$\Rightarrow \frac{|a-1|}{\sqrt{a^2+b^2}} = \frac{k}{4}$

So $ax + by = 1$ touches the circle $(x-1)^2 + y^2 = \left(\frac{k}{4}\right)^2$.

50. 5.44
Range (5.43 – 5.45)

Sol. $y = \left(1+\frac{1}{x}\right)^x \Rightarrow \ln y = x\ln\left(1+\frac{1}{x}\right)$

$\lim_{x\to\infty} x\ell n\left(1+\frac{1}{x}\right) = 1.$

51. 2.72
Range (2.71 – 2.73)

Sol. Let $\ell = \left\{f\left(\frac{1}{n}\right).f\left(\frac{2}{n}\right)f\left(\frac{3}{n}\right).....f\left(\frac{n}{n}\right)\right\}^{2/n}$

$\lim_{n\to\infty} \ell n(\ell) = \lim_{n\to\infty} \frac{2\left[\ell n f\left(\frac{1}{n}\right) + \ell n f\left(\frac{2}{n}\right) + \ell n f\left(\frac{3}{n}\right) + ... + \ell n f\left(\frac{n}{n}\right)\right]}{n}$

$= 2\int_0^1 \ell n f(x)\,dx$ (definite integral limit)

$= 2\int_0^1 \ell n\left(1+\frac{1}{x}\right)^x dx = 2\int_0^1 x\,\ell n\left(1+\frac{1}{x}\right)dx$

$= 2\left[\ell n\left(1+\frac{1}{x}\right).\frac{x^2}{2}\right]_0^1 - 2\int_0^1 \frac{\left(-\frac{1}{x^2}\right)}{1+\frac{1}{x}}.\frac{x^2}{2}dx$ (By ports)

$= \ell n 2 + \int_0^1 \frac{x}{x+1}dx = 1.$

So, $\ell = e^1 = e = 2.72$.